

Clinical Risk Assessment Report: Patient 25

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 37.5% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.70x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET), which elevated the patient's absolute risk by +6.0%. Biologically, this feature points to continuous, elongated bands of high-density or highly active tumor tissue.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: 0.81 [Top 12%] (+6.0%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: 2.17 [Top 2%] (+3.8%)**
Reflects the most dense solid components or calcifications within the primary tumor lesion.

Tumor Pixel Correlation (CT) (GLCM_Correlation.CT) **Value: 1.72 [Top 2%] (+3.5%)**
Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: 0.23 [Top 7%] (+3.4%)**
Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.75 [Avg (73th %ile)] (+3.0%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 0.07 [Avg (52th %ile)] (+2.9%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

3. Protective / Mitigating Factors (Reducing Risk)

Minimum Tumor Density (CONVENTIONAL_HUmin) **Value: -2.76 [Bottom 1%] (-2.6%)**
Reflects the least dense areas within the tumor on CT, often corresponding to cavitation, necrosis, or cystic degeneration.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: 3.82 [Top 1%] (-2.5%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Machine Learning Feature Attribution (SHAP Decision Path)

